PCC UNIT-4

1. Explain why Multiple Access Techniques are needed.

A.

1. **Efficient Spectrum Utilization**
   - Radio spectrum is limited.
   - Multiple access techniques (FDMA, TDMA, CDMA, OFDMA) allow many users to share the same frequency band without interfering with each other.
   - This maximizes the use of available bandwidth.
2. **Support for Multiple Users**
   - Cellular systems must serve thousands or millions of users simultaneously.
   - Multiple access ensures that each user gets a fair share of resources (frequency, time, or code).
3. **Quality of Service (QoS)**
   - By separating users, interference is reduced.
   - This improves call clarity, data throughput, and reliability.
   - Essential for maintaining acceptable voice and data quality.
4. **Mobility Management**
   - Users move across cells while maintaining active connections.
   - Multiple access techniques enable seamless handoffs and resource allocation during mobility.
5. **Capacity Expansion**
   - Techniques like CDMA and OFDMA allow overlapping use of spectrum with minimal interference.
   - This increases the number of users the system can support.
   - Critical for modern systems like LTE and 5G.
6. **Flexibility for Different Services**
   - Voice, SMS, and high-speed data have different requirements.
   - Multiple access techniques allow networks to allocate resources dynamically depending on service type.
7. **Interference Control**
   - By assigning distinct frequencies, time slots, or codes, interference between users is minimized.
   - This ensures stable communication even in dense urban environments.

2. Define capacity in case of TDMA, FDMA, CDMA

A. **Capacity in FDMA (Frequency Division Multiple Access)**

- **Definition:** Capacity is determined by the number of frequency channels available.
- **Formula (conceptual):**

$$\text{Capacity} = \frac{\text{Total available bandwidth}}{\text{Bandwidth per channel}}$$

- **Key Point:** Each user gets a dedicated frequency channel.
- **Limitation:** Once all frequency slots are occupied, no more users can be served.
- **Impact:** Capacity is strictly limited by spectrum size and channel spacing.

Capacity in TDMA (Time Division Multiple Access)

- **Definition:** Capacity is determined by dividing each frequency channel into multiple time slots.
- **Formula (conceptual):**

$$\text{Capacity} = \text{Number of frequency channels} \times \text{Number of time slots per channel}$$

- **Key Point:** Multiple users share the same frequency but transmit in different time slots.
- **Limitation:** Capacity depends on synchronization and guard times between slots.
- **Impact:** More efficient than FDMA, as it allows multiple users per frequency.

Capacity in CDMA (Code Division Multiple Access)

- **Definition:** Capacity is determined by the number of users that can be supported simultaneously using unique codes, while maintaining acceptable signal quality.
- **Formula (conceptual):**

$$\text{Capacity} \approx \frac{W}{R} \times \frac{1}{E_b/N_0}$$

where:

- W= system bandwidth
- R= data rate per user
- E_b/N_0= required energy per bit to noise density ratio
- **Key Point:** All users share the same frequency band, separated by codes.
- **Limitation:** Capacity is interference-limited; as more users join, noise rises and quality drops.
- **Impact:** Flexible capacity, but depends heavily on power control and interference management.

3. Explain space division multiple access techniques.

A. **Space Division Multiple Access (SDMA)**

SDMA is a multiple access technique where users are separated in **space** using directional antennas or beamforming. Instead of dividing spectrum by frequency, time, or code, the system allocates different spatial paths to different users.

1. **Directional Antennas / Smart Antennas**
   - The base station uses antenna arrays to form narrow beams.
   - Each beam covers a specific spatial direction.
   - Different users in different directions can use the same frequency and time slot without interference.
2. **Beamforming**

    - Signals are steered toward the intended user.
    - Interference from other directions is minimized.
    - This allows spatial reuse of frequencies.
3. **Spatial Reuse**
    - Multiple users can share the same frequency band and time slot if they are separated spatially.
    - Increases system capacity without requiring extra bandwidth.

Advantages

- **Higher Capacity:** Same frequency/time resources reused in different spatial directions.
- **Reduced Interference:** Beams are focused, minimizing cross-talk between users.
- **Better Coverage:** Adaptive antennas can track users and maintain strong signals.
- **Supports Advanced Systems:** Essential for LTE and 5G, where massive MIMO and beamforming are key.

4. What is frequency hopping?

A. **Frequency Hopping**

- **Frequency hopping** is a technique in wireless communication where the carrier frequency of a signal is changed rapidly according to a predefined sequence.
- The transmitter and receiver both follow the same hopping pattern, so communication remains synchronized.
- It is a form of **spread spectrum** technique.

1. The available frequency band is divided into multiple channels.
2. The transmitter changes (or “hops”) from one channel to another at fixed intervals.
3. The receiver, knowing the hopping sequence, tunes to the same frequencies at the same times.
4. Communication continues seamlessly, even though the carrier frequency keeps changing.

Types

- **Fast Frequency Hopping (FFH):** Frequency changes several times during transmission of one data bit.
- **Slow Frequency Hopping (SFH):** Frequency changes after transmitting multiple bits on one frequency.

Advantages

- **Interference Resistance:** If one frequency is noisy, the system quickly hops to another.
- **Security:** Makes interception and jamming harder, since the hopping sequence is not easily predictable.
- **Multipath Reduction:** Reduces fading effects by spreading transmission across different frequencies.

- **Efficient Spectrum Use:** Multiple users can share the same band with different hopping sequences.

5. What is the near-far problem in CDMA, and how does it impact network performance?

A. **Near–Far Problem in CDMA**

- **Definition:** In CDMA (Code Division Multiple Access), all users share the same frequency band simultaneously, separated only by unique codes. The **near–far problem** occurs when a mobile user close to the base station transmits with high power, while another user farther away transmits with weaker power. The strong signal from the "near" user can overwhelm the weaker signal from the "far" user, making it difficult for the receiver to correctly decode the far user's transmission.
- CDMA relies on correlation properties of codes to separate users.
- If one signal is much stronger than another, the weaker signal gets buried in interference.
- Unlike FDMA/TDMA, where users are separated by frequency or time, CDMA users overlap in both frequency and time, so power imbalance directly affects decoding.

Impact on Network Performance

1. **Reduced Capacity:**
   - If near–far imbalance is not controlled, fewer users can be supported simultaneously.
   - The system becomes interference-limited.
2. **Poor Voice/Data Quality:**
   - Far users experience degraded signal quality, leading to dropped calls or low throughput.
   - Voice clarity and reliability suffer.
3. **Unfair Resource Usage:**
   - Near users dominate the channel, while far users struggle to maintain connections.
   - This reduces overall system efficiency.
4. **Increased Dropped Call Rate (DCR):**
   - Far users are more likely to lose connections, raising the dropped call rate.

6. Discuss about the features of TDMA.

A. **Features of TDMA (Time Division Multiple Access)**

1. **Time Slot Division**
   - Each frequency channel is divided into multiple time slots.
   - Multiple users share the same frequency but transmit in different time slots.
2. **Efficient Spectrum Use**
   - Allows more users per frequency compared to FDMA.

    - Improves capacity by multiplexing in time.
3. **Synchronization Requirement**
    - Precise timing is needed so that users transmit only in their assigned slots.
    - Guard times are inserted to prevent overlap between slots.
4. **Digital Transmission**
    - TDMA is well suited for digital signals.
    - Used in 2G GSM systems and early digital cellular networks.
5. **Flexibility**
    - Time slots can be dynamically allocated depending on traffic demand.
    - Supports both voice and data services.
6. **Reduced Interference**
    - Since users transmit in separate time slots, co-channel interference is minimized.
    - However, strict synchronization is essential.
7. **Capacity Expansion**
    - Capacity is determined by:

$$\text{Capacity} = \text{Number of frequency channels} \times \text{Number of time slots per channel}$$

- More time slots → more users supported.

8. **Power Efficiency**
    - Mobile devices transmit only during their assigned slot.
    - Reduces average power consumption, extending battery life.

7. Discuss about the features of CDMA.

A. CDMA (Code Division Multiple Access) is a digital multiple access technique where multiple users share the same frequency band simultaneously, distinguished by unique codes. Its key features include soft capacity, resistance to interference, and improved call quality through spread spectrum technology.

Features of CDMA

1. **Shared Frequency Band**
    - All users transmit over the same frequency spectrum.
    - Separation is achieved using unique spreading codes rather than time or frequency slots.
2. **Spread Spectrum Technology**
    - Signals are spread over a wide bandwidth using pseudo-random codes.
    - Provides robustness against interference and eavesdropping.
3. **Soft Capacity Limit**
    - Unlike FDMA/TDMA, CDMA does not have a fixed user limit.
    - Capacity depends on interference levels: more users increase the noise floor, gradually reducing quality.
4. **Near–Far Problem & Power Control**
    - Strong signals from nearby users can overpower weaker signals from distant users.
    - CDMA systems require strict **power control** to balance received signal

strengths.

5. **Soft Handoff**
    - Mobile devices can connect to multiple base stations simultaneously.
    - Ensures smoother call continuity and reduces dropped calls compared to hard handoff.
6. **Improved Security**
    - Spreading codes make signals harder to intercept or jam.
    - Provides inherent privacy and resistance to unauthorized access.
7. **Better Voice Quality**
    - Low interference and wideband spreading improve clarity.
    - Multipath signals can be combined constructively, reducing fading effects.
8. **Flexible Resource Allocation**
    - Since all users share the same band, resources can be dynamically managed.
    - Supports both voice and data services efficiently.

8. Compare TDMA, FDMA, SDMA, CDMA.

A.

Technique	**Principle**	**Resource Division**	**Key Features**	**Merits**	**Demerits**
FDMA	Frequency Division Multiple Access	Spectrum divided into fixed frequency channels	Each user gets a dedicated frequency; guard bands prevent overlap	Simple, continuous transmission, low latency	Limited capacity, inefficient spectrum use, rigid allocation
TDMA	Time Division Multiple Access	Each frequency channel divided into time slots	Users share same frequency but transmit in different slots; requires synchronization	Efficient spectrum use, supports multiple users per channel, power saving	Synchronization complexity, guard times reduce efficiency, latency due to slot waiting
CDMA	Code Division Multiple	All users share same frequency band, separated by unique codes	Spread spectrum; signals distinguishe	High capacity, better voice	Near–far problem, interference-limited, complex

	Access		d by correlation; requires power control	quality, soft handoff, secure	receivers
SDMA	Space Division Multiple Access	Users separated spatially using directional antennas/beamforming	Smart antennas form beams toward users; spatial reuse of frequencies	High capacity, reduced interference, efficient reuse	Requires advanced antenna technology, complex implementation

9. Explain Frequency Division Multiple Access techniques and mention its merits and demerit

A. **Frequency Division Multiple Access (FDMA)**

Definition: FDMA is a multiple access technique where the available frequency band is divided into multiple channels, and each user is assigned a unique frequency channel for the duration of the call or data session.

- Each user → separate frequency.
- Channels are separated by **guard bands** to avoid interference.
- Used in **1G analog systems** and still forms the basis for many radio systems.

FDMA Works

1. The total spectrum is split into fixed frequency slots.
2. Each slot is allocated to one user at a time.
3. Communication continues on that frequency until the call ends.
4. Guard bands ensure signals don't overlap between adjacent channels.

Merits of FDMA

- **Simplicity:** Easy to implement with analog technology.
- **Dedicated Channel:** Each user has a fixed frequency, ensuring stable communication.
- **Low Latency:** No need for synchronization in time (unlike TDMA).
- **Continuous Transmission:** Suitable for voice calls, since users transmit continuously.

Demerits of FDMA

- **Limited Capacity:** Number of users is restricted by available frequency slots.
- **Inefficient Spectrum Use:** Guard bands waste part of the spectrum.
- **Rigid Allocation:** Channels are fixed; unused slots cannot be shared dynamically.
- **Poor Scalability:** Not suitable for modern systems with high user density.
- **Interference Sensitivity:** Adjacent channel interference can occur if guard bands are insufficient.

10. What are the types of multiple access techniques? Explain detail.

A. Multiple access techniques are methods that allow many users to share the limited radio spectrum simultaneously without interfering with each other. The main types are FDMA, TDMA, CDMA, and SDMA/OFDMA, each using a different principle (frequency, time, code, or space).

1. Frequency Division Multiple Access (FDMA)

- **Principle:** Spectrum is divided into distinct frequency channels; each user gets a dedicated frequency.
- **Operation:** Users transmit continuously on their assigned frequency. Guard bands are used to prevent overlap.
- **Features:**
  - Simple to implement (used in 1G analog systems).
  - Low latency since transmission is continuous.
  - Limited capacity due to fixed frequency slots and wasted guard bands.
- **Merit/Demerit:** Stable communication but inefficient spectrum use.

2. Time Division Multiple Access (TDMA)

- **Principle:** Each frequency channel is divided into time slots; users share the same frequency but transmit in different slots.
- **Operation:** Requires strict synchronization; guard times prevent overlap.
- **Features:**
  - More efficient than FDMA (used in 2G GSM).
  - Supports multiple users per frequency.
  - Requires precise timing control.
- **Merit/Demerit:** Efficient spectrum use, but synchronization complexity is high.

3. Code Division Multiple Access (CDMA)

- **Principle:** All users share the same frequency band simultaneously, separated by unique spreading codes.
- **Operation:** Uses spread spectrum; signals are spread over wide bandwidth and distinguished by correlation.
- **Features:**
  - Soft capacity (number of users depends on interference levels).
  - Requires strict power control to avoid near–far problem.
  - Supports soft handoff for seamless mobility.
- **Merit/Demerit:** High capacity and security, but interference-limited and complex.

4. Space Division Multiple Access (SDMA)

- **Principle:** Users are separated spatially using directional antennas or beamforming.
- **Operation:** Base station forms narrow beams toward each user, allowing spatial reuse of frequencies.
- **Features:**
  - Increases capacity by reusing same frequency/time resources in different directions.

- Reduces interference with smart antennas.
- **Merit/Demerit:** Very efficient, but requires advanced antenna technology.

5. Orthogonal Frequency Division Multiple Access (OFDMA)

- **Principle:** Extension of FDMA used in 4G LTE and 5G. Spectrum is divided into many orthogonal subcarriers.
- **Operation:** Each user is assigned a set of subcarriers dynamically.
- **Features:**
  - High spectral efficiency.
  - Flexible resource allocation for voice, video, and data.
  - Robust against multipath fading.
- **Merit/Demerit:** Excellent for broadband data, but computationally complex.

11. Explain Time Division Multiple Access techniques and mention its merits and demerit

A. **Time Division Multiple Access (TDMA) – Technique**

- **Principle:** TDMA divides each frequency channel into multiple **time slots**. Each user is assigned a specific slot, and they transmit only during that slot.
- **Operation:**
  - Multiple users share the same frequency but at different times.
  - Guard times are inserted between slots to prevent overlap.
  - Requires strict synchronization between base station and mobiles.
- **Usage:** Widely used in **2G GSM systems** and early digital cellular networks.

Merits of TDMA

1. **Efficient Spectrum Use:** Multiple users can share one frequency channel.
2. **Flexibility:** Time slots can be dynamically allocated depending on traffic demand.
3. **Reduced Interference:** Users transmit in separate slots, minimizing co-channel interference.
4. **Digital Compatibility:** Well suited for digital signals (voice + data).
5. **Power Efficiency:** Mobiles transmit only during their slot, saving battery life.

Demerits of TDMA

1. **Synchronization Complexity:** Requires precise timing control; guard times reduce efficiency.
2. **Limited Data Rates:** Each user gets only a fraction of the channel capacity.
3. **Latency:** Users must wait for their slot, which can introduce small delays.
4. **Rigid Slot Allocation:** If a slot is unused, it cannot be shared dynamically with others.
5. **Scalability Issues:** Not ideal for very high-density networks compared to CDMA/OFDMA.

12. Discuss about Code Division Multiple Access techniques and mention its merits and demerit

A. **Code Division Multiple Access (CDMA) – Technique**

- **Principle:** CDMA allows multiple users to share the same frequency band simultaneously.

- **Operation:**
    - Each user is assigned a unique **spreading code**.
    - Signals are spread over a wide bandwidth using these codes.
    - At the receiver, correlation with the correct code extracts the intended signal while others appear as noise.
- **Key Features:**
    - Based on **spread spectrum technology**.
    - Supports **soft handoff** (mobile connected to multiple base stations at once).
    - Requires strict **power control** to avoid the near–far problem.
    - Provides **soft capacity** (capacity depends on interference levels, not fixed slots).

Merits of CDMA

1. **High Capacity:** More users can be supported compared to FDMA/TDMA, since all share the same band.
2. **Better Voice Quality:** Wideband spreading reduces interference and fading.
3. **Soft Handoff:** Seamless mobility with reduced call drops.
4. **Security & Privacy:** Signals are difficult to intercept due to spreading codes.
5. **Flexible Resource Allocation:** Dynamic sharing of bandwidth for voice and data.
6. **Resistance to Interference:** Spread spectrum makes CDMA robust against narrowband interference.

Demerits of CDMA

1. **Near–Far Problem:** Strong signals from nearby users can overpower weaker ones unless power control is applied.
2. **Complexity:** Requires sophisticated receivers and correlation techniques.
3. **Interference-Limited:** As more users join, noise rises, reducing quality.
4. **Strict Power Control Needed:** Continuous adjustment of mobile transmit power is essential.
5. **Soft Capacity Drawback:** No fixed user limit; quality degrades gradually with load.